

ORIGINAL ARTICLE

A randomized study of computerized working memory training and effects on functioning in everyday life for patients with brain injury

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Abstract

Background: Working memory (WM) problems influence most activities of daily living. The aim was to evaluate if computerized working memory training after brain injury has a significant effect on functioning in daily life.

Method: Outpatients with WM deficits, aged 22–63 years, were randomized to either intervention group (IG, $n = 20$) or control group (CG, $n = 18$) and received 5 weeks standard rehabilitation. The IG also received WM training with the Cogmed QM training program. Assessments were made before (A1), immediately (A2) and 3 months (A3) after intervention. After follow-up, the CG was offered the computerized training and assessed after this (A4; $n = 8$). Assessments included the WAIS-III Digit span reversed, Fatigue Impact Scale (FIS), Assessment of Motor and Process Skills (AMPS), Rivermead Behavioural Memory Test-II (RBMT-II) and a WM questionnaire.

Results: The IG improved on digit span and FIS, A1–A2, and significantly more than the CG on the WM questionnaire, A1–A3. Both groups improved in AMPS motor skill and the AMPS process skill score tended towards significant improvement in the IG, from A1–A3. After training (A3–A4), the CG improved in digit span and RBMT-II.

Conclusion: The WM training seems to have a generalized effect on functional activity and lessens fatigue.

Keywords

Fatigue, functional activity, generalization, rehabilitation, working memory

History

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Introduction

Working memory (WM) refers to the ability to pick up and hold information over a brief period of time, for as long as is needed to perform an action [1]. Working memory is important for several cognitive processes [2]. Problems with WM manifest as reduced attention and planning and can affect persons in most occupations [3]. Therefore, deficits in WM, which are common after acquired brain injury, have great implications for functioning in daily life [4]. There is some evidence that working memory is linked to key learning outcomes as information needs to enter working memory before it can be stored into long-term memory [5]. The larger the capacity of working memory for certain stimuli the faster these materials can be learned [6]. Many activities in daily life depend on short-term memory and in addition the individuals need active maintenance and processing of information, for example when one has to step out of routine activity like driving somewhere else than the usual trip to work [7]. Ability to perform everyday memory activities is crucial for the possibilities to lead an independent life. Working memory impairment can result in a failure to remember instructions and the inability to complete activities.

The rehabilitation of cognitive impairments relies on remedial and adaptive approaches [8, 9]. Recent studies have found that WM can be improved by a remedial approach to training [10, 11]. The computerized Cogmed WM training program devised by Klingberg et al. [12] (RoboMemo for children and Cogmed QM for adults, earlier called ReMemo) has been used for several different diagnostic groups yielding good results, including children and adults with Attention Deficit Hyperactive Disorder, adolescents with borderline intellectual disabilities, children with cochlear implants and individuals with stroke and other brain injuries [13–18]. Most of these studies have also shown favourable results on non-trained tasks.

Transfer from trained to non-trained tasks could be explained by the way in which prior learning affects new learning [19]. Transfer of training to everyday life does not always occur. Transfer is supposed to occur by two mechanisms: (1) low road transfer or reflexive transfer, which is produced by routine iteration in a nearby context to the original learning; and (2) high-road transfer or mindful transfer, which is a conscious abstraction and seeking for an association to other situations [20]. In rehabilitation transfer is a key function that always needs to be addressed and evaluated before starting an intervention [19]. To ensure a transfer effect a 'multi context approach' described by Toglia et al. [21] is recommended to use. If transfer is successful the

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outcome should be observed as improved ability in different and for the patient important activities. With an adaptive approach to training, the likelihood of transfer depends on the similarity of the task and the context, as well as the learning ability of the client [19, 22, 23]. With a remedial approach to training, the hypothesis is that the capacity improves if the training is successful and does not depend on context or learning abilities. The training with the Cogmed QM might correspond to a remedial approach in that the tasks in the computerized training program are based on intensive repetition on a for the patient adapted and challenging performance level [18]. However, the transfer of gains in body functions, such as WM, to everyday life is not easily measured in a standardized manner, as performance of most activities in everyday life depends on several cognitive functions in interaction with environmental and personal factors [19, 24]. It is of great importance for rehabilitation services that methods used in training are evidence-based and efficient in improving the patient's functioning and participation in social life. Åkerlund et al. [25] reported that Cogmed QM training improved general cognitive function and psychic health. Based on data from the same sample as Åkerlund et al. [25], the aim of this study was to evaluate if Cogmed QM training has a significant impact on activity and participation.

Method

This study took place in Gothenburg, Sweden, between March 2008 and December 2010. It was a randomized and controlled study designed to explore the feasibility of the computerized training program Cogmed QM for patients of working age (18–65 years) in the post-acute phase after a brain injury. All patients admitted for outpatient rehabilitation were considered for participation. Some of these were allocated to other projects. From the remaining patients those identified with WM deficits at enrolment, were invited to participate in the study [25]. Eligibility was determined by a score of ≤ 5 digits/blocks forwards and/or ≤ 4 digits/blocks reversed on the WAIS-III Digit span and WAIS-III-NI Span board. Patients who were not able to communicate or were contraindicated to participate due to medical reasons according to the physician's health assessment (i.e. pronounced fatigue, pain or depression) were excluded. All patients received written and verbal information about the study and those who agreed to participate submitted written informed consent before the baseline assessment.

Demographic data were collected from the medical chart. The participants were randomized into two groups: the intervention group (IG) and the control group (CG; Figure 1). The CG was offered rehabilitation in accordance

with the usual routines at the clinic, based on their rehabilitation needs. Computerized cognitive training was excluded. The IG was also offered rehabilitation in accordance with the usual routines at the clinic, but with additional training with the Cogmed QM training program. The program was used according to the guidelines, i.e. 30–45 minutes per session, 5 days a week for 5 weeks (25 sessions). The program includes a series of exercises for visuospatial and verbal WM. The program is set to continuously adapt to the progress and cognitive capacity of the participant and gives immediate feedback to the participant. The training also included personal and individual feedback once a week by a specially trained coach.

Assessments

To explore the impact of the training on several levels, assessments were made according to the International Classification of Functioning on the levels of body function and activity/participation using the WAIS-III Digit span reversed, the Assessment of Motor and Process Skills (AMPS), the Rivermead Behavioural Memory Test-II (RBMT-II), the Fatigue Impact Scale (FIS) and a custom-made questionnaire of everyday situations that place great demand on working memory (WM questionnaire).

The WM questionnaire consists of 20 questions rated on a visual analogue scale (VAS), from 'not at all' (0) to 'almost always' (100) and reported as an index. This questionnaire was specially designed for this project to explore WM in more complex activity. To find out its potential to measure executive functions it was correlated (Spearman) to the DEX, reported in another study [25], with a result of; $\rho = 0.674$, $p = 0.000^{**}$.

To assess WM, the level of body function, the WAIS-III Digit span reversed was used as the analyses of the neuropsychological tests in the study by Åkerlund et al. [25] showed that this was the most sensitive and suitable variable to use for assessing change in WM.

The AMPS is an instrument for observation and evaluation of occupational performance in two domains of activities of daily living: motor skills and process skills [26, 27]. It is a standardized instrument and a valid and reliable measure. The result is given in logits. A clinically relevant change in AMPS score is 0.3–0.5 logits [28]. The AMPS was administered by a clinical occupational therapists trained, calibrated and certificated in the administration of the AMPS.

The RBMT-II is a short, reliable, ecological valid test used to assess everyday memory problems [29–31]. It is designed to predict everyday memory problems in people with acquired

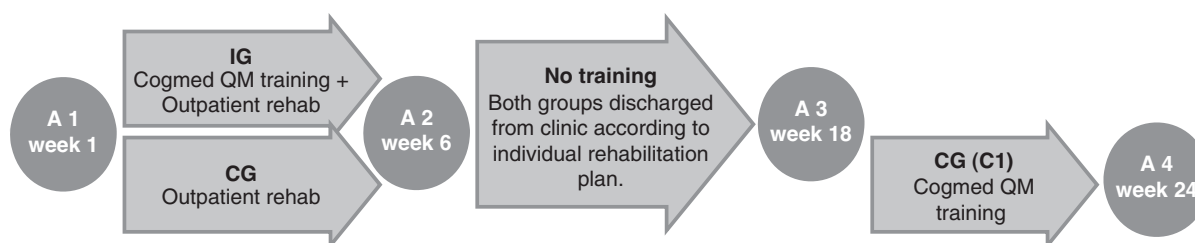


Figure 1. Schematic showing study design and assessment time points for the control group (CG) and intervention group (IG). Assessments were made before (A1), immediately (A2) and 3 months (A3) after the 5-week intervention period. For CG participants who continued with a period of Cogmed QM training after A3 (C1), a fourth assessment was made after training (A4).

brain injury and to monitor their change over time [32]. This study focused on the importance of working memory for the performance of more complex activities which demands also storage and long-term memory, as in everyday memorial activities. As it was suggested that the RBMT probably is the best known and most frequently used instrument for assessing everyday memory [33], it was chosen for the purpose to capture if improvements in working memory transfer also to everyday activities.

The FIS was developed to improve understanding of the effects of fatigue on quality-of-life. The FIS items reflect the perceived impact of fatigue on cognitive, physical and psychosocial functioning. The version of the FIS with 40 questions [34] that has been translated and validated for a Swedish population was used. The questions are rated from 0 (no problem) to 4 (very much a problem) [34, 35].

The Dysexecutive Questionnaire (DEX) is a self-report questionnaire measuring experienced dysexecutive problems in emotion, personality, motivation, behaviour and cognition to assess problems in goal-directed behaviour [36]. The results of this questionnaire are reported by Åkerlund et al. [25] in the other part of this study. In this study it was only used for validating the outcome of the WM questionnaire.

Assessments were made at baseline (A1), after the 5 week training programme (A2) and at follow-up 3 months later (A3; Figure 1). After the 3-month follow-up assessment, the CG was offered the Cogmed QM training program and, if they accepted, participants were assessed again after the 5 week training period (A4; Figure 1).

Analyses

Descriptive data are presented as mean and standard deviation (SD) or median and range. The WM questionnaire was analysed as an index and according to the percentage of participants that had rated themselves >50 on the VAS line, i.e. experiencing problems. For all instruments, the change in score between different assessment time points, including pre- to post-intervention (A1–A2) and pre- and post-intervention compared to follow-up (A1–A3 and A2–A3), was computed for both groups. For the participants in the CG who received the Cogmed QM training, the change in score between pre- and post-training (A3–A4) was also computed. Within each group, change over time was analysed using the non-parametric Wilcoxon signed rank test for paired samples. Change over time was also compared across groups (CG and IG) using the non-parametric Mann Whitney U-test for independent samples.

Due to the small sample size all data from the analyses are presented with both statistical significance and effect size in order to provide information supporting that the sample size was good enough to detect a non-random difference as well as to show that the findings could be clinical relevant. The level of significance was set to $p < 0.05$. The statistical package SPSS (version 15.0) was used. The SPSS does not provide effect size statistics for non-parametric analyses, however, this has been calculated according to the recommendations in the SPSS survival manual, third edition [37]. Interpretation of the effect size (r) was made according to Cohen [38], $r = 0.1$ small effect, $r = 0.3$ medium effect and $r = 0.5$ large effect.

In the preceding analysis of data from the same sample [25], the CG was split into two sub-groups: CG participants who received the Cogmed QM training program after the 3-month follow-up (group C1, $n = 8$) and CG participants who declined the Cogmed QM training program (group C2, $n = 10$). The sub-groups differed in subjective perception of problems with depression where the C1 reported more depressive symptoms than C2 [25]. In this study, these sub-groups were used in the presentation of the percentage of participants that perceived problems with WM demanding situations, i.e. score of >50 on the WM questionnaire.

Ethical considerations

The study was performed within the ordinary clinical work at the department as a development effort with permission from the head of the department and has, therefore, not been considered by an ethical committee. Ethical consideration was made throughout to secure participant safety and integrity according to professional and hospital ethical rules.

Results

Of 331 patients admitted to outpatient rehabilitation, 128 patients were allocated to other projects. All remaining 203 patients were screened for the study. The inclusion criteria of ≤ 5 digits/blocks forwards or ≤ 4 digits/blocks reversed in the WAIS-III Digit span and WAIS-III-NI Span board were met by 121 patients. Twenty-four of these were excluded due to communication problems (e.g. aphasia, unable to speak Swedish) or not being fit for testing. Fifty patients declined participation and the remaining 47 patients accepted and were included in the study. The remaining 45 participants were assigned to the IG ($n = 25$) or the CG ($n = 20$). During the study, there was a further seven drop-outs for various reasons. Data are presented from the 20 participants in the IG and 18 participants in the CG who completed the study. Eight participants in the CG accepted the offer of the Cogmed QM training after the 3-month follow-up period and these patients form sub-group C1.

The sample consisted of 16 men (42%) and 22 women (58%), with a mean age of 51 years (SD = 11 years, range = 22–63 years). The median time since injury was 27 weeks (range = 12–500 weeks). The cause of WM problems was stroke ($n = 28$, 74%), traumatic brain injury ($n = 5$, 13%) or other ($n = 5$, 13%). The majority, 63.8%, were fully conscious and 8.5 were unconscious at admittance to the emergency ward. No significant differences were found between the IG and CG regarding gender, age, time since injury, consciousness level, educational level, diagnosis or scaled scores for working memory at A1. Åkerlund et al. [25] also reported that no significant differences between the IG and CG were found at baseline assessment (A1) in any of the psychometric tests (The Digit span, Span board, Working Memory sub-scale of WAIS-III NI, Barrow Neurological Institute Screen of Higher Cortical Functions, HADS and DEX).

Demographic characteristics, WAIS-III Digit span reversed score, AMPS scores, RBMT-II score, FIS score and WM questionnaire index at A1 were similar in the IG and the CG. The mean WM questionnaire index at A1 was 45, SD = 15

(median = 41, range = 27–72), in the IG and 42, SD = 23 (median = 49, range = 2–82), in the CG (Table I). Both groups reported problems with WM in several areas of daily life. The areas where most participants reported difficulties were: remembering items from a recipe or notes, remembering if invoices had been paid, finding the car in a parking garage, keeping track of a complex conversation, remembering without writing down and performing mental calculations (Figure 2). Unfortunately, due to missing data on this questionnaire, it was not possible to follow the change over time other than between A1 and A3. Table I shows the percentage of participants in the IG, C1 and C2 that scored >50 on the WM questionnaire, indicating WM problems.

From A1 to A2, the IG improved significantly on the Digit span reversed score ($p = 0.003$, effect size $r = -0.48$) and the FIS score ($p = 0.038$, $r = -0.33$), but the CG made no significant improvements on any measure (Figure 3). From A2 to A3, the IG improved significantly on the AMPS motor skill score ($p = 0.038$, $r = -0.33$), but the CG made no significant improvements on any measure (Figure 3). From A1 to A3, both groups improved significantly on the AMPS motor skill score ($p = 0.016$, $r = -0.38$ and $p = 0.013$, $r = -0.41$ for IG and CG, respectively). The median change in the IG was 0.32 logits and in the CG 0.48 logits, indicating

that both groups made clinically relevant changes (>0.30 logits). The AMPS process skill score also somewhat increased in the IG, but this did not reach significance ($p = 0.084$, $r = -0.27$, 0.29 logits). From A1 to A3, the IG improved significantly on the Digit span reversed score ($p = 0.000$, $r = -0.56$). The proportion of participants in the IG that had an index of >50 on the WM questionnaire decreased significantly ($p = 0.006$, $r = -0.44$) between A1 and A3. During this period the CG did not improve significantly on the WM questionnaire. Of the participants in the CG who decided to participate in the Cogmed QM training program (C1, $n = 8$), the Digit span reversed score ($p = 0.039$, $r = -0.52$) and the RBMT score ($p = 0.042$, $r = -0.51$) significantly improved from A3 to A4 (Figure 3) and the Digit span reversed score ($p = 0.041$, $r = 0.51$) significantly improved from A1 to A4.

The change in the Digit span reversed score from A1 to A3 was greater in the IG than the CG ($p = 0.003$, $r = -0.48$). At A2, the IG already had shown a tendency for greater improvement, but this was not significant ($p = 0.059$, $r = -0.31$). The decrease in the proportion of participants who had an index of >50 on the WM questionnaire from A1 to A3 was greater in the IG than in the CG ($p = 0.020$, $r = -0.38$).

Discussion

This study was performed to determine if improvements in WM after a computerized training program transferred to improvements in functioning in daily life in participants with acquired brain injury. However, functioning in daily life is dependent on many aspects and it is difficult to capture the contribution of improvement in a single function such as WM. Thereby, although it may appear that the evidence for functional improvement is scarce, the results all indicate the positive conclusion that computerized training can improve reaching rehabilitation goals concerning everyday functioning

Table I. Percentage and number (*n*) of participants in the intervention group (IG) and the control group that did (C1) and did not (C2) receive the Cogmed QM training that had an index above 50 on the working memory questionnaire before the intervention period (A1), 3 months after the end of the intervention period (A3) and after the additional Cogmed QM training (A4) for C1.

Group	A1	A3	A4
IG	35% (<i>n</i> = 20)	14% (<i>n</i> = 14)	
C1	50% (<i>n</i> = 8)	60% (<i>n</i> = 5)	33% (<i>n</i> = 3)
C2	20% (<i>n</i> = 10)	33% (<i>n</i> = 6)	

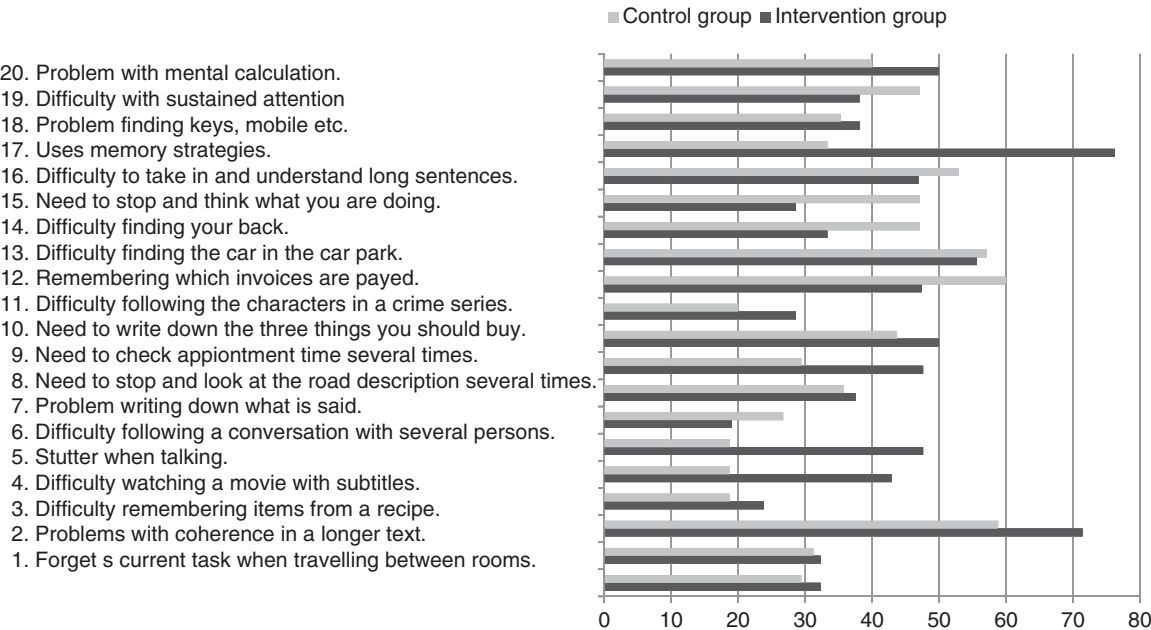


Figure 2. The percentage of the participants in the intervention group (dark grey) and the control group (light grey) that rated themselves >50 on a visual analogue scale of 0 (not at all) to 100 (almost always) for each question of the working memory questionnaire at baseline (A1).

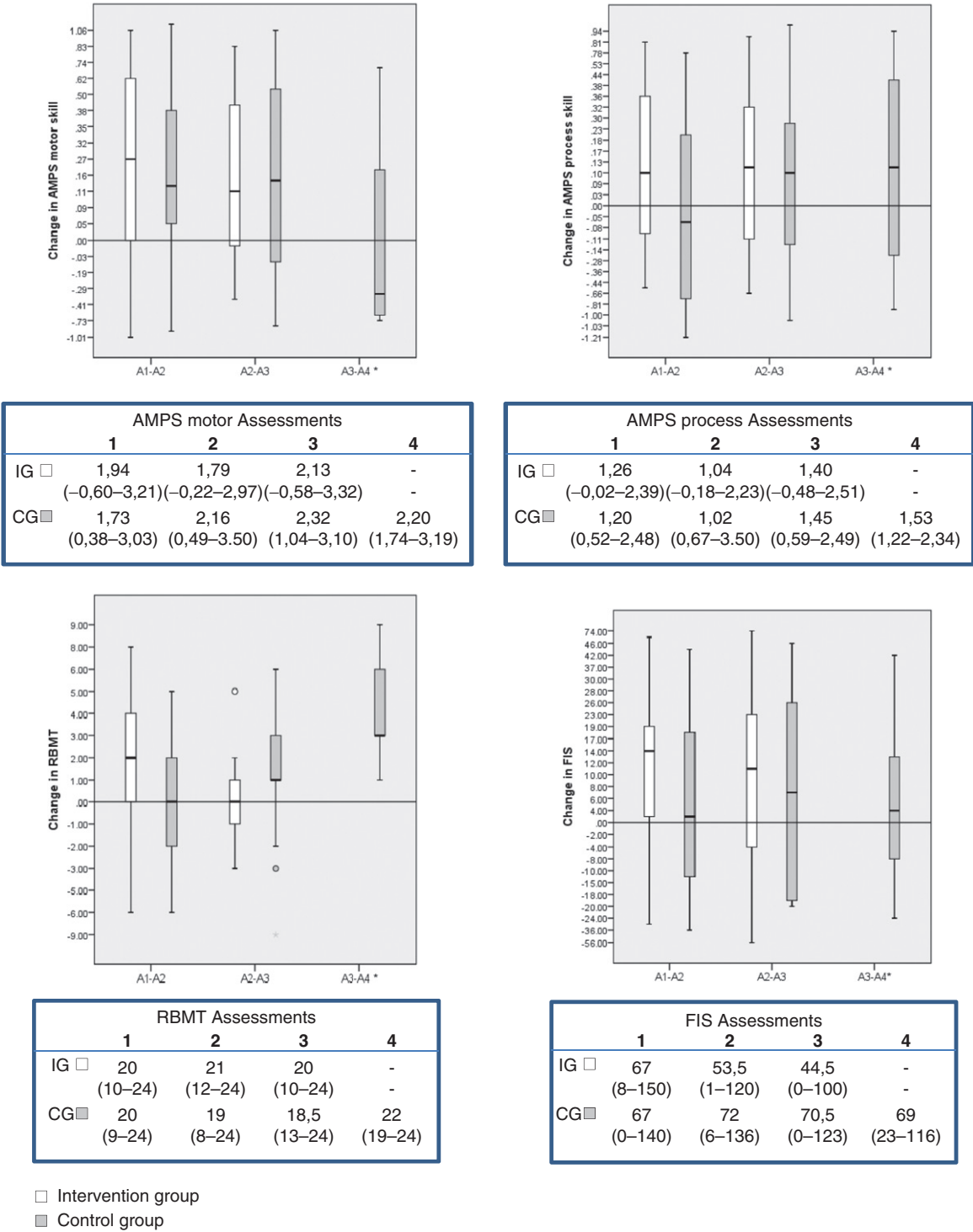


Figure 3. Boxplots illustrating the change in Assessment of Motor and Process Skill (AMPS) motor skill score, AMPS process skill score, Rivermead Behavioural Memory Test-II (RBMT) score and Fatigue Impact Scale (FIS) score from A1 to A2 and A2 to A3 in the intervention group (white) and the control group (grey) and from A3 to A4 in the control group who received the Cogmed QM training after A3. The hinges of the box represent the 25th and 75th percentile the whiskers represent 95% of cases. In the frame below the boxplot, the median and range at each assessment are presented.

after brain injury. The improvements that the IG showed from pre- to immediately post-intervention were mainly on a direct measure of WM function (Digit span). However, from pre- to 3 months post-intervention, there was also an improvement in AMPS motor skill score, the WM questionnaire index and the AMPS process skill score. The need to assimilate the gain in WM may explain that improvements in occupational performance not were presented until later.

The decrease in fatigue after training was an additional positive improvement that also could have implications for functioning in real life.

Eight individuals in the CG also received Cogmed QM training after the initial study period. The results after the training in these participants support the suggestion that the training effect can be generalized to different aspects of activity, as there were improvements in everyday memory

(as indicated by RBMT-II score) and a subjective perception of fewer problems in WM demanding situations.

This study has some limitations that may have influenced the outcomes and the extent to which the results can be generalized. As in many clinical studies, there were problems in recruiting a large sample. In addition, there were several drop-outs during the study period. These circumstances could, however, be seen as a common problem in studies with brain injured participants, who often experience fatigue as a limiting factor after the injury. Previous studies have reported similar problems and have adapted the training program in order to facilitate participation [16, 17]. This study is the largest to date, as far as we know, of Cogmed QM training in adults after acquired brain injury. An advantage over previous studies is that this study included a CG that also received rehabilitation. Primarily for ethical reasons, a period of Cogmed QM training was provided for the CG after the follow-up period, and the effects of this training were also evaluated. However, this study did not perform an assessment of the IG at the A4 time point, which would have allowed for comparison of the change in both groups as a cross-over design. Less than half the participants in the CG accepted the offer of training, and it is suspected that it would have been difficult to persuade the participants in the IG to return for a fourth assessment, as they found the assessments quite tiring. As many analyses of the data were made, it may have been appropriate to adjust the level of significance to avoid Type I errors. However, because the study may be somewhat under-powered it was decided not to adjust the level of significance and instead tried to be cautious in drawing conclusions. Another limitation is that one only could analyse changes in the WM questionnaire between the A1 and A3 time points. However, the effect size analyses support clinical evidence of the results.

The study was part of clinical work that involved several therapists and was performed over a long period of time; therefore, it was hard to ensure that all data were collected. The WM questionnaire was the authors' own design, as there is a lack of feasible instruments with focus on activity and participation. However, the WM questionnaire was correlated to the DEX to validate the statements of executive functions in the custom-made WM questionnaire according to the statements of executive functions in the DEX. Both these instruments contain allegations that require WM to function. The results correlated significantly, indicating the WM questionnaire to have a substantial content according to executive functions which are required in complex activities of daily living. In recently reported studies of the same training program, the Cognitive Failure Questionnaire [39] was used and this may have been a valid alternative for use in the current study [17, 18].

Åkerlund et al. [25] reported results from the same sample and analysed the CG in sub-groups to explore differences between those participants who decided to train the Cogmed QM program after follow-up (A3) and those who did not. There were differences between the groups in the ratings of depression symptoms [25]. In this study, differences were found with regard to how the groups rated their experience of WM problems. Both studies, therefore, found that the sub-groups differed on self-rating scales that reflect participants' own perceptions of feelings and ability, where the C1 reported

more problems. This might influence self-image and underlines the need for individualization of rehabilitation programmes.

The WM questionnaire showed that most participants perceived themselves to have WM difficulties that could have a great impact on daily life. The percentage of participants that scored highly on this questionnaire was lower at follow-up than prior to training in the IG, but augmented at follow-up in the CG, although the sub-groups differed in level of problems as there was a much higher percentage in the C1 compared to the C2 that scored highly. The difference in direction of the change between the IG and the CG might be attributed to the Cogmed QM training and a feeling in the IG that they had done something to contribute to their improvement. Johansson and Tornmalm [17] made conclusions from self-ratings and a qualitative analysis of diaries in their study of effects of WM training that supports this explanation.

In this study, RBMT-II was used to predict memory problems of daily life. The RBMT-II only improved significantly for the C1 group, between A3 and A4 after their Cogmed QM training. The fact that both groups (IG and CG) were close to the maximum score may have contributed to the lack of improvement. This may be an indication that the RBMT-II instrument is not sensitive enough to reflect real-life difficulties, as both groups reported great difficulties in daily life on the WM questionnaire. RBMT-II mainly assesses a lower level of memory of everyday activities. This also implicates some limitation according to generalizing to the ability to function in daily life where often much more complex activities have to be memorized and planned for to be performed. However, having difficulties to remember and perform the tasks described in RBMT-II suggests even greater problems in real life which are supported by the subjective experiences of the patients.

A study by McNab et al. [11] found that WM, a key function for human cognition and dependent on adequate dopamine neurotransmission, can improve from WM training and they concluded that training seems to be associated with changes in the density of cortical dopamine D1 receptors. This implies that WM training influences brain plasticity, which could help to explain the transfer of WM training effects to other domains such as activity. Fatigue is also dependent on dopamine transmission [11] and this may explain the decrease in fatigue for the IG after Cogmed QM training. In this case, the C1 group also should have shown this decrease after receiving the training; however, the small sample size may have prevented a similar finding in the C1 group. Alternatively, the lack of change in fatigue in the C1 group might indicate that there is more benefit on fatigue if the training is performed earlier in the rehabilitation process. The training program is demanding and intense and, therefore, this study had many patients who declined participation or dropped out. An update from the program provider, Cogmed, means that the program is now internet-based and training can take place in the patient's home. This will save time and energy for the patient and hopefully make the program suitable for more patients. Perhaps this internet-based program will allow for a better mix with complimentary adaptive training and thereby result in a more evident gain in everyday life.

For the assessment of motor and process skill, the AMPS was chosen as an objective measure of occupational performance in daily activities. These activities are complex and require not only WM but also many other skills and would probably require a substantial improvement in WM to manifest as improvement of motor and process skills. From A1 to A3, both groups made significant improvements on the AMPS motor skill score. An improvement of 0.3–0.5 logits on the AMPS has been found to be clinically meaningful in terms of improved occupational performance [28] and the median improvement seen in both groups fell within this range. There was a relation between WM and motor skill score, perhaps because improved WM decreases the energy load and allows more attention to be given to motor actions, resulting in a better quality of motor performance. As shown by Åkerlund et al. [25] in the same sample, both groups made significant improvements on the aggregated WM sub-scale of the WAIS-NI, which is in accordance with the improvement in AMPS motor skill score. The WM gain for both groups is explained by the fact that both groups received rehabilitative interventions. This study used the WAIS-III Digit span reversed as a measure of WM and, on this test, only the IG made significant improvements between A1 and A3. However, the reason for improvements in both groups on the AMPS motor skill score and the aggregated WM sub-scale of the WAIS-NI [25] could be that both these measures are a combination of different functions, whereas the Digit span reversed specifically represents a single function that, by itself, is not easily manifested on the more generalized measure of the AMPS. However, there was a significant correlation between improvement on the Digit span reversed and the AMPS process skill score in the IG, which might be seen as a transfer effect of the improvements in WM to improvements in performance due to the Cogmed QM training. The process skill includes actions to logically sequence steps, to select appropriate tools and materials and to adapt the performance when problems are encountered. In addition, the training in the IG seemed to generate a positive trend of decrease in fatigue, which also may be of great importance to occupational performance. With the limitations of the small sample in mind, these results, although some are only tentative, are of importance for the evidence that support the use of the Cogmed QM training as a treatment method to achieve better occupational performance after brain injury.

Conclusions

There are three major findings with regard to an indication of transfer of improvements in WM to improvements in performance. Both the IG and the CG made significant improvements in the aggregated WM score from WAIS-III and the AMPS motor skill score. The IG additionally showed a significant improvement in the more specific measure for WM, the Digit span reversed, a significant decrease in fatigue and had a tendency for the improvement in the AMPS process skill score to correlate with the change in the Digit span reversed. A large proportion of the IG made improvements that can be viewed as clinically relevant. After the Cogmed QM training, the C1 group also showed improvement on the Digit span reversed as well as in everyday memory, as

indicated by the RBMT score. The findings of better functioning after training were also confirmed by the subjective experience reported in the WM questionnaire. In conclusion, although the results must be interpreted cautiously, in light of the small sample size, these results give some indications of positive transfer at the activity level after computerized WM training in the sub-acute phase after acquired brain injury. This should be followed by future studies with larger samples to obtain stronger evidence.

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Declaration of interest

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

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